FUTURE HEALTH | A GLOBAL INITIATIVE BY ABU DHABI | H2S HACK2SKILL

Future Health Digital Innovation Challenge

From signals to solutions.

Co-creator Microsoft

Co-creator inception42

Co-creator Mohamed bin Zayed University of Artificial Intelligence

Team Name: Design Master Solutions

Team Size: 1

Problem Statement: [Population Health] Closing the Screening-to-Care Loop.

Solution: Loop Closer

Team members*
[Name, country, role, organization, age confirmation]

S.No	Name	Country	Role	Organization	Age confirmation
1	Alan De Bortolo	Brazil	Founder, lead developer	Design Master Solutions LLC (USA)	Confirmed: 18 or older
2					
3					
4					
5					

Team Capability Summary*

Maximum 1,000 characters

Design Master Solutions LLC is a Massachusetts software company founded in 2019. It designs, codes, hosts and supports its own products, with web, mobile and backend work done in house.

The founder also builds All4Clinic, a clinic management system in Brazil used by 24 paying clinics. Its appointment confirmation reads free-text patient replies and opens a task for whoever does not answer, the same pattern this challenge needs. No All4Clinic code or data is used here.

Gap we state openly: the team has no clinician. Before any pilot, the clinical lead of the screening program must own the pathway, the service levels and the patient wording.

Architecture or workflow diagram*

Case pathway. One accountable owner and one due date for every step.

Result issued	Patient notified	Understanding confirmed	Confirmatory test	Referral sent	Referral accepted	Appointment attended	Treatment or surveillance plan	Closed with evidence

INPUT
HL7 v2 ORU^R01 or FHIR R4 from the lab or program. One case per accession number. Built and tested on sample messages

CASE ENGINE
service level by urgency, owner per step, flags: unassigned, overdue, referral rejected, consult waiting, escalate

HUMAN CONTROL
Coordinator worklist shows only cases that need a person. Only a clinician can close, and only with evidence

PATIENT
Plain message in Arabic or English, never a diagnosis. Reply 1 understood, 2 talk to a nurse

CONSULTATION
Patient-to-clinician and clinician-to-clinician questions on the case, with their own service level

MEASURES
Notified, understood, referral completion, days to specialist, unresolved, why cases stalled

Built as pure functions with no I/O (public/loop.mjs), so the same logic runs in the browser and in the tests. No AI decides anything clinical in this prototype.

Target users and intended outcome*

Screening coordinators and care navigators: start the day with the short list of cases that need a person, not a spreadsheet of every result.

Primary care doctors and specialists: see who owns the next step across organizations, accept or reject a referral, and answer questions on the case.

Patients: get a clear next step in their language, confirm they understood, and reach a nurse with one reply.

Program leaders: see where cases wait and why, with documented closure for each actionable result.

Intended outcome: every actionable screening result is communicated, confirmed and converted into timely care, with a named owner and documented closure. No result stalls without somebody seeing it.

How will success be measured?*

Prototype scenario on the same 60 simulated cases: a stall stays silent, or the worklist surfaces it and coordinators resolve 80% of exceptions two days after the breach. Model assumptions, not a measured outcome.

Measure	Silent	Worklist
Closed with evidence	19	29
Stalled, nobody acting	23	7
Patient notified	93%	98%
Understanding confirmed	89%	95%
Referrals completed	67%	73%

In a pilot, against the program baseline:

Time to notification, confirmation, consultation and care

Referral and appointment completion

Unresolved actionable results and reasons for drop-off

Share of closures with documented evidence

Patient understanding and confidence, by language

Median days to specialist does not move in this scenario: recovered cases arrive later, so it needs real data.

Evidence supporting the approach*

Citations, studies, benchmarks, or prior testing.

Failure is common. Callen et al., J Gen Intern Med 2012;27:1334-48: systematic review of 19 studies; tests not followed up ranged from 6.8% to 62% for laboratory tests and 1.0% to 35.7% for radiology, with missed cancer diagnoses among the impacts.

Delay matters. Corley et al., JAMA 2017;317:1631-41: 70,124 patients with a positive FIT; follow-up after 10 months was associated with higher colorectal cancer risk and more advanced stage than follow-up at 8 to 30 days.

Follow-up needs a system. Doubeni et al., CA Cancer J Clin 2018;68:199-216: PROSPR systematic review and recommendations on timely follow-up of positive cancer screening results. Selby et al., J Med Screen 2021;28:51-53: international survey of interventions that ensure follow-up of positive FIT.

Prior testing. 30 automated tests pass. FHIR output checked with the official HL7 validator and round-tripped on a public FHIR R4 server (17 resources created, read back, no duplicates on resend). No clinical testing yet.

Data sources and legal right to use them*

List source, type, owner, sensitivity, permissions, and whether personal data is used.

Source	Type	Owner	Personal data	Permission
Simulated cohort (60 cases, fixed seed)	Synthetic	Design Master Solutions	No	Own work, MIT license
Staff names in the demo	Fictional	Design Master Solutions	No	Invented for the demo
Patient messages (EN, AR)	Text written for the prototype	Design Master Solutions	No	Needs clinical and language review before use
Pilot: screening results and referrals	Health data, high sensitivity	Screening program and providers	Yes	Not used. Requires program agreement, health authority approval and patient consent basis

Risks and safeguards*

Privacy; security; bias; explainability; misuse; human oversight; failure modes.

Privacy and security: no patient data in the prototype, and the public API refuses any message not marked as test data. A pilot needs UAE hosting, role-based access, an audit log and encryption.

Bias: rule-based, no model scoring patients. We report measures by language and program to catch unequal drop-off.

Explainability: every flag comes from a visible rule: owner missing, due date passed, referral rejected.

Misuse: patient text always says it is not a diagnosis. Wording is reviewed by the program.

Human oversight: urgent cases with any flag go to the top. Only a clinician closes, and only with evidence.

Failure modes: built: silent-feed alarm, reconciliation against the lab list, corrected results flag the open case, unknown or unreadable results go to a person. In a pilot, unsent messages must surface as exceptions. Service levels are illustrative until the program sets them.

Hardware requirements

Required if wearable, sensor, connected device, or other hardware is there. Include sourcing and safety.

Not applicable. Loop Closer is software only and needs no wearable, sensor or connected device.

Path to pilot and scale*

Stakeholders, integration, regulatory needs, timeline assumptions, and sustainability.

Stakeholders: Department of Health Abu Dhabi, screening program leads, laboratories, primary care, specialist services, a named clinical lead.

Integration: built and tested on sample data: results in by HL7 v2 ORU^R01 or FHIR R4; case state out as an idempotent FHIR R4 transaction (Task, ServiceRequest, Communication, Encounter, CarePlan, Provenance). Next: a production lab interface and Malaffi, the Abu Dhabi health information exchange. Scale: over 10,000 messages per second on one laptop core (docs/bench.txt).

Regulatory: UAE Federal Law No. 2 of 2019 on ICT in health fields (health data stays in the UAE unless an exception applies; long retention). Formal legal review before any pilot.

Timeline assumption: 1 to 2 months to agree pathway and service levels with one program; 3 months of integration and security review; 3 to 6 months of shadow pilot on one screening program, then a measured comparison with its baseline.

Sustainability: annual subscription per program, justified by measured closure. Pricing to be validated in the pilot.

Pre-existing IP and third-party materials*

Identify ownership, licenses, open-source components, and restrictions.

Ownership: all code was written for this challenge from 30 October 2026, in a new repository owned by Design Master Solutions LLC, MIT license: github.com/DesignMaster-Solutions/loop-closer.

Pre-existing IP: none. No code or data from All4Clinic or other company products.

Open-source and third-party: no runtime dependencies. Node.js test runner for tests, HL7 FHIR Validator and the public HAPI test server to check output, Wrangler to deploy, Cloudflare Workers to host the demo.

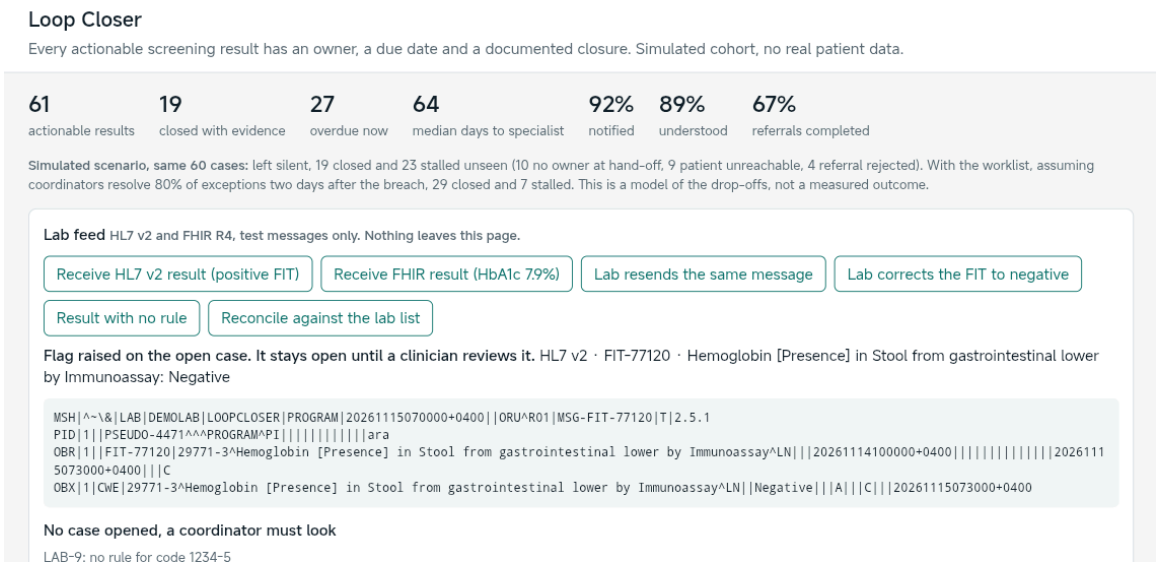
AI assistance: built with an AI coding assistant, reviewed and tested by the team.

Restrictions: none known in the prototype. Clinical wording and pilot data would belong to the program.

Conflict of interest declaration*

Disclose relationships with organizers, partners, or judges.

None. No member of our team has a conflict of interest to declare.



Live demo: loop-closer.designmaster.workers.dev

Repository: github.com/DesignMaster-Solutions/loop-closer

Limitations: simulated cohort; service levels and scenario rates are assumptions; Arabic text needs review; FHIR and HL7 v2 tested on sample data only, no production connection; demo changes stay in the browser.

Additional Slide

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